Assignment 2: Automated S3 Bucket Cleanup Using AWS Lambda and Boto3

Objective

The objective of this assignment is to automate the cleanup of files in an Amazon S3 bucket that are older than 30 days using AWS Lambda and Boto3. This helps in cost optimization and automated storage management.

AWS Services Used

Amazon S3

AWS Lambda

IAM

Boto3 (Python SDK)

Amazon CloudWatch

Implementation Steps

1. Created an S3 bucket named assignment2-s3-cleanup-rohan-vijay-mangate.
2. Uploaded sample files into the S3 bucket for testing.
3. Created an IAM role with AmazonS3FullAccess and attached it to Lambda.
4. Created a Lambda function using Python 3.12.
5. Implemented Python code using Boto3 to delete objects older than 30 days.
6. Manually tested the Lambda function using the AWS console.

Lambda Function Code Explanation

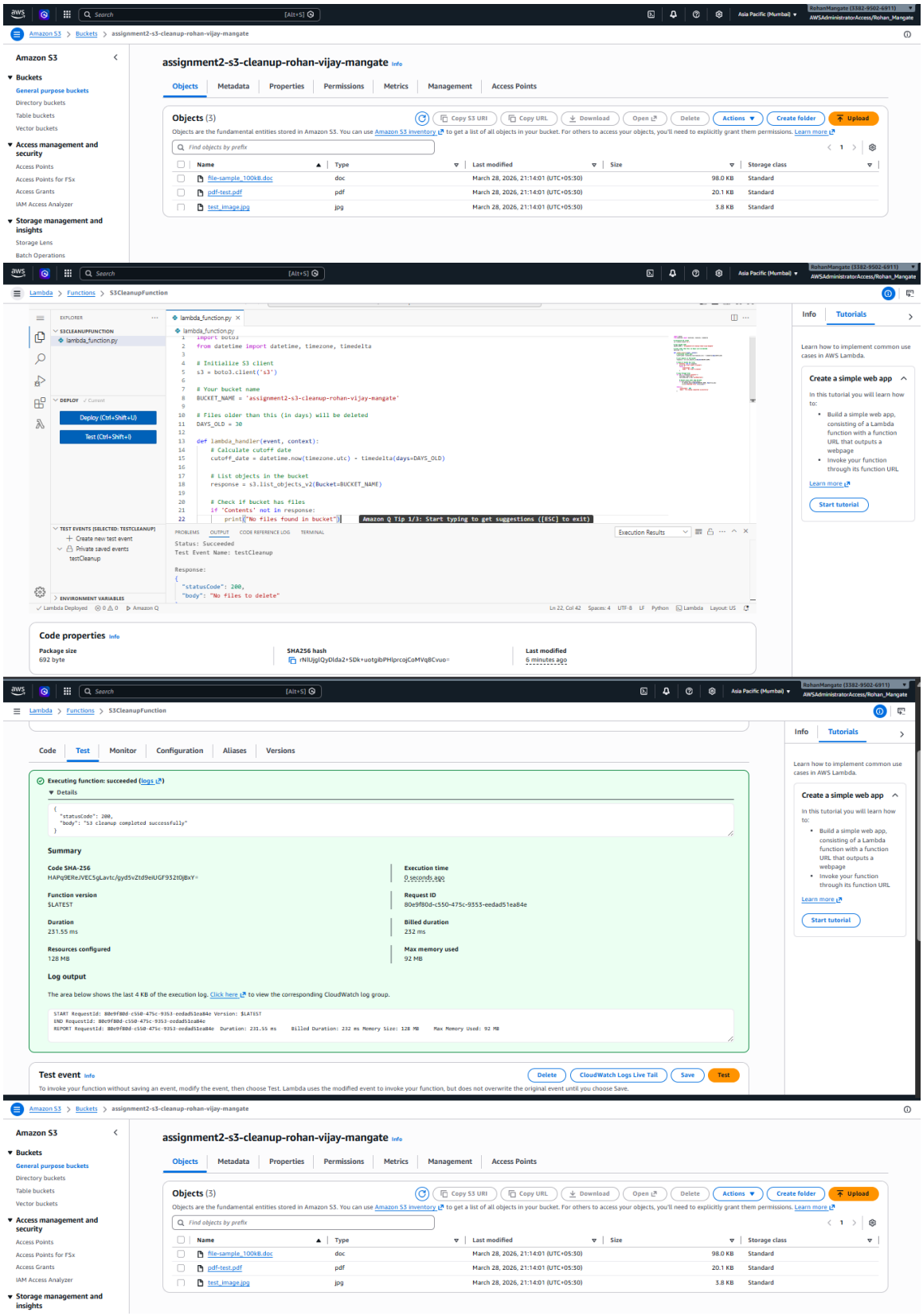
The Lambda function initializes an S3 client using Boto3, lists all objects in the bucket, checks their LastModified timestamp, and deletes files that are older than 30 days. Execution logs are printed to CloudWatch for verification.

Testing and Validation

The Lambda function was manually triggered using a test event. The execution completed successfully with a status code of 200, confirming that the cleanup logic works as expected.

Screenshots

The following screenshots are included for validation:
1. S3 bucket showing uploaded files before cleanup.
2. Lambda function Python code.
3. Lambda test execution result showing successful run.
4. CloudWatch logs and/or S3 bucket after cleanup.



Conclusion

This assignment demonstrates how AWS Lambda and Boto3 can be used to automate S3 bucket maintenance. The solution successfully fulfills the assignment objectives by implementing automated cleanup logic and validating it through manual testing.